YONG JING WEN A23CS0202

Lab 7

Discussion Questions

1. **Explain how Lab Activity 07 extends the cafe application developed in Lab Activity 06.**

   Lab Activity 07 extends upon the previous cafe application by adding the form validation, API integration, data retrieval and Vue components. In the Lab Activity 06, the application primarily used Vue to handle basic food ordering functionality. On the other hand, the Lab Activity 07 system became more complex and capable of validating user input, submitting orders to a simulated backend using a JSON server, retrieving saved orders using Axios and displaying a list of saved orders.

2. **Explain the role of props in the order-form and order-list components.**

   Props are used to pass data from the parent Vue application to child components. Besides, props are used to receive the food list from the parent application in the order-drive component. This allows the form to display available food options in a dropdown list.

   In the order list component, props are used to receive the order list from the parent application. This allows the component to display all saved cafe orders in a table.

3. **Explain how emits are used to send order data from the child component to the parent app.**

   When a child component needs to send data back to its parent component, it can use the ‘emits method.

   In this lab, the order-form component collects the customer's name, selected food, quantity and total price. After the form validates successfully, the child component will use `this.$emit('submit-order', newOrder);` which helps to send the new order data to the parent application. The parent application listens for this event which is `@submit-order="submitOrder"`. The parent application then runs the submitOrder() method to save the order via Axios.

4. **Describe how form validation is handled before an order is submitted.**

Form validation is handled in the ‘handleSubmit()’ method of the order form component. Before submitting an order, the system checks whether the customer name or selected food is null or the quantity is less than 0. After that, the system will display “Please complete all fields correctly” if the input is invalid.

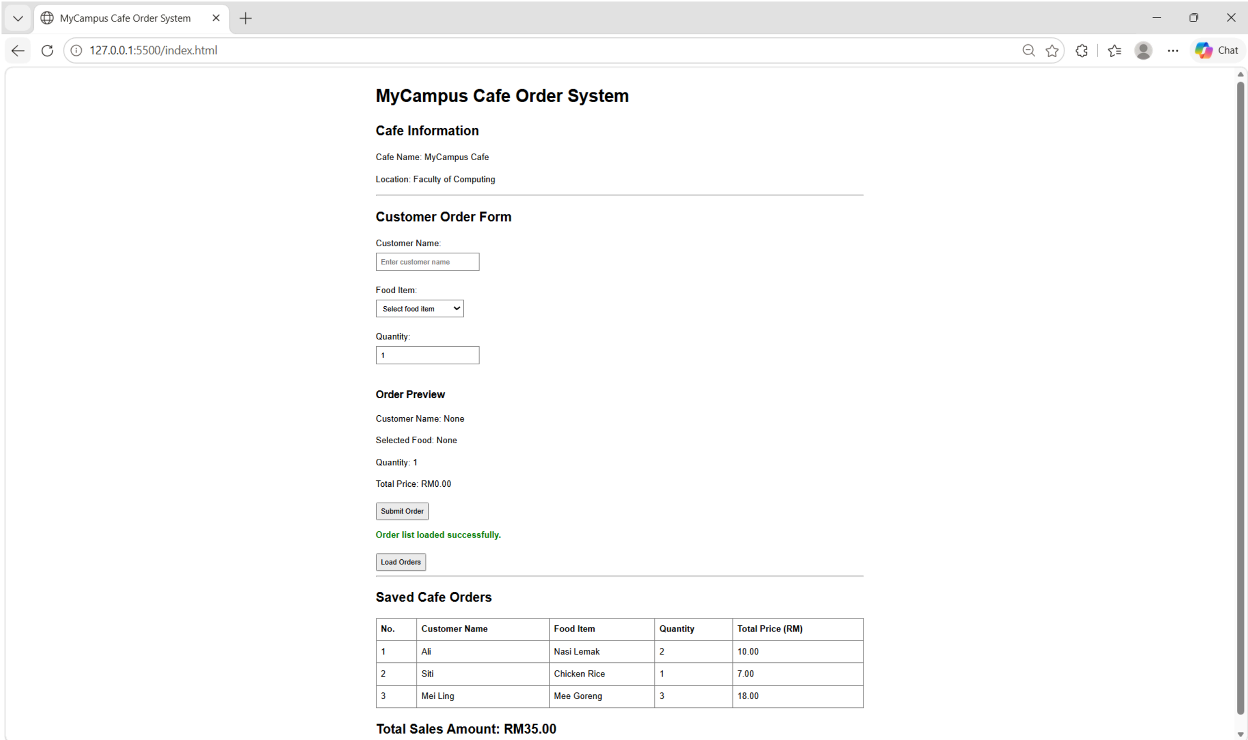
5. **Explain the role of Axios in submitOrder() and fetchOrders().**

   Axios is used to communicate with the simulated backend API. In the submitOrder() function, Axios is used to send new order data to the server via a POST request. In the FetchOrder(), Axios is used to retrieve saved order records from the server using GET requests.

6. **Explain why the component-based version is more readable than placing the whole interface logic in one large block.**

   Component-based versions are more readable because the code is broken down into smaller parts. The order form component handles only customer input and validation, while the order list component displays saved orders.

   This makes the application easier to understand, maintain and update. Each component has a clearly defined responsibility rather than cramming all the HTML, form logic, validation and order display logic into one large block of code. This also makes the code more reusable and organized.



MyCampus Cafe Order System

127.0.0.1:5500/index.html

MyCampus Cafe Order System

Cafe Information

Cafe Name: MyCampus Cafe

Location: Faculty of Computing

Customer Order Form

Customer Name:

Enter customer name

Food Item:

Select food item

Quantity:

1

Order Preview

Customer Name: None

Selected Food: None

Quantity: 1

Total Price: RM0.00

Submit Order

Order list loaded successfully.

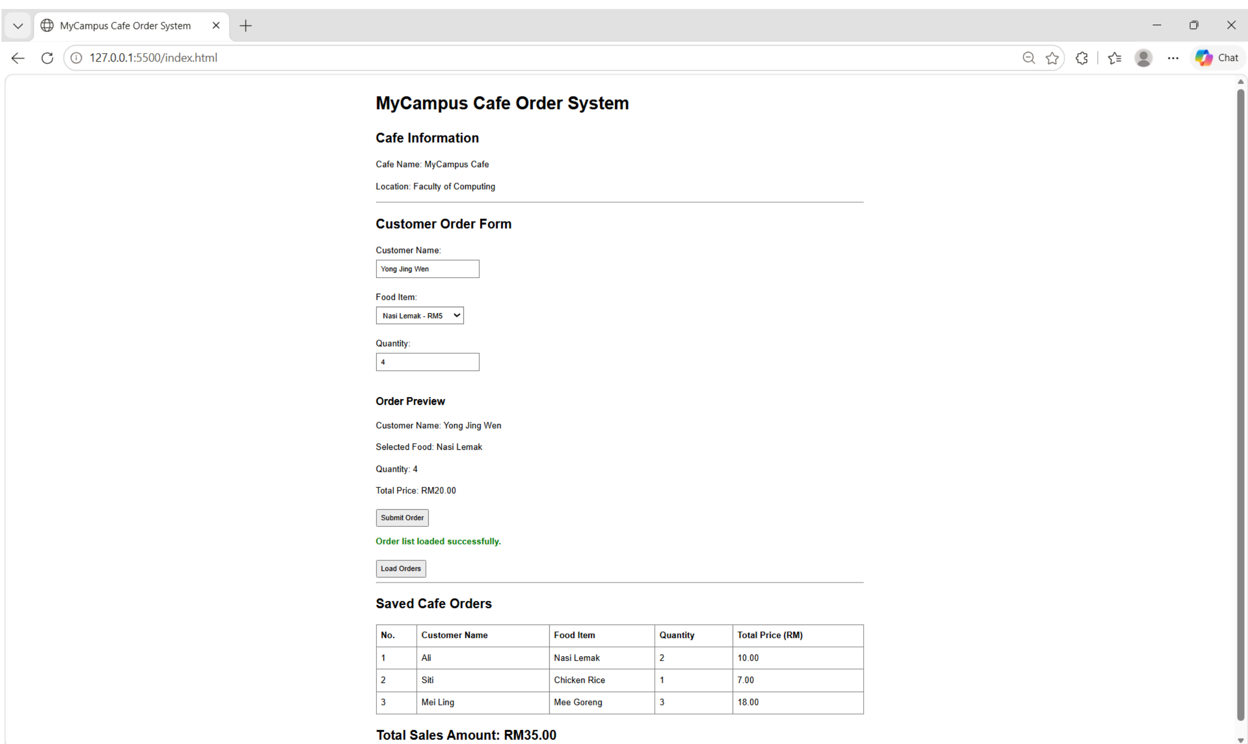
Load Orders

Saved Cafe Orders

No.	Customer Name	Food Item	Quantity	Total Price (RM)
1	Ali	Nasi Lemak	2	10.00
2	Siti	Chicken Rice	1	7.00
3	Mei Ling	Mee Goreng	3	18.00

Total Sales Amount: RM35.00

Figure 1.1 MyCampus Cafe Order System form



MyCampus Cafe Order System

127.0.0.1:5500/index.html

MyCampus Cafe Order System

Cafe Information

Cafe Name: MyCampus Cafe

Location: Faculty of Computing

Customer Order Form

Customer Name:

Yong Jing Wen

Food Item:

Nasi Lemak - RM5

Quantity:

4

Order Preview

Customer Name: Yong Jing Wen

Selected Food: Nasi Lemak

Quantity: 4

Total Price: RM20.00

Submit Order

Order list loaded successfully.

Load Orders

Saved Cafe Orders

No.	Customer Name	Food Item	Quantity	Total Price (RM)
1	Ali	Nasi Lemak	2	10.00
2	Siti	Chicken Rice	1	7.00
3	Mei Ling	Mee Goreng	3	18.00

Total Sales Amount: RM35.00

Figure 1.2: MyCampus Cafe Order System form after the customer order details have been filled in

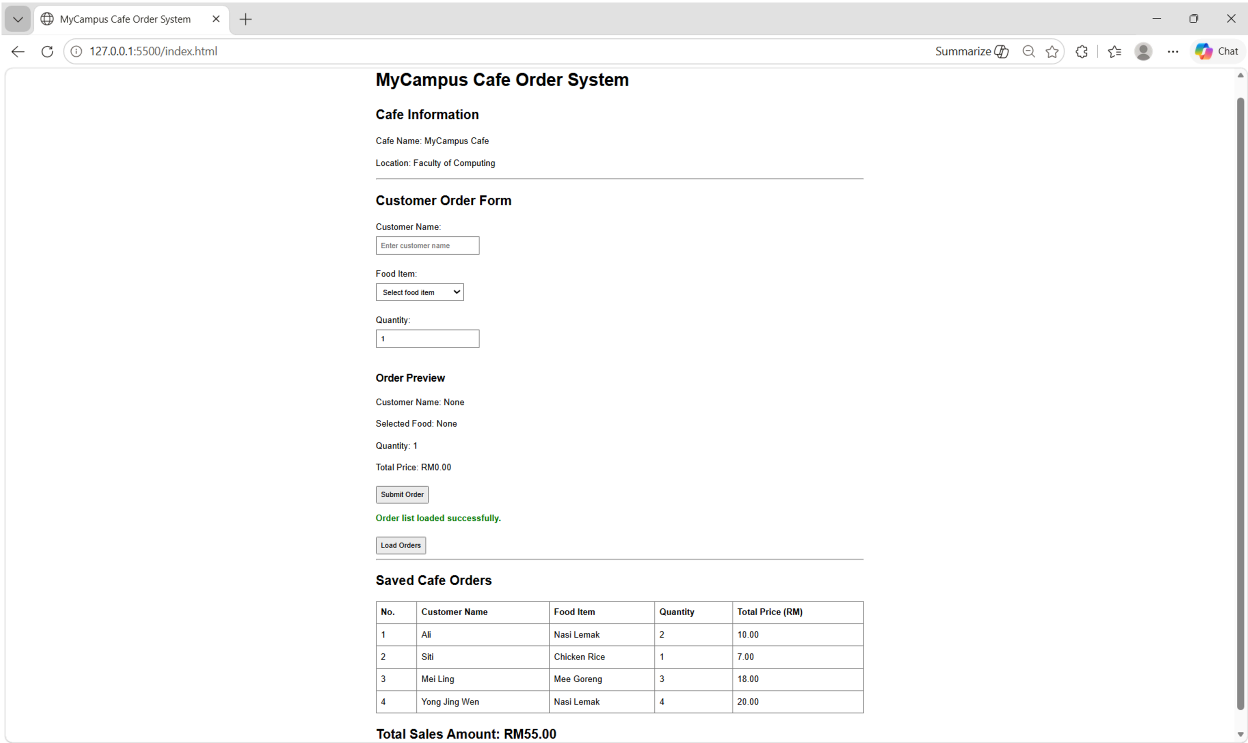


Figure 1.3: MyCampus Cafe Order System form after the customer submit their order

Lab 8

Lab Questions

1. **Explain the role of router-link and router-view in this application.**
   In this application, router-link is used to create navigation links for single-page application (SPA) pages, such as "Home," "Add Order," and "View Orders." It allows users to navigate between pages without refreshing the entire browser window.

   Besides, router-view is used to display the selected route component. For example, when the user clicks "Add Order” , the addOrderPage component will appear in the router-view. When the user clicks "View Orders”, the ViewOrdersPage component will appear.

2. **Explain how the shared store supports simple state management across routes.**
   By using Vue.reactive(), it stores common data such as food items, orders, and messages. Due to the data store being shared, different route components can access the same data. For instance,, the "Add Order" page uses a list of food items from the data store and the "View Orders" page uses a list of orders from the same data store.
   This helps to prevent repeating the same data in every component.

3. **State one reason why createWebHashHistory() is suitable for this CDN-based SPA lab.**
   The createWebHashHistory() function works because it uses the # symbol in the URL which makes the application more efficient without refreshing the browser when switching the pages.This is particularly effective for CDN-based Vue applications because it doesn't require additional server-side routing configuration.

4. **Explain why fetchOrders() is placed outside the route components in this design.**
   The fetchOrders() function is placed outside the route component so that it can be reused across different parts of the application. Therefore, the same function no need to be repeatedly written. For example, after a user submits a new order on the "Add Order" page, the fetchOrders() function is called to refresh the order list. On the "View Orders" page, the function is also called to load saved orders from a JSON server.

5. **Differentiate between route navigation and component rendering in this application.**

   Route navigation refers to navigating between different page routes using Vue Router. For example, when a user clicks "Home," "Add Order" or "View Orders", the URL changes and Vue Router determines which page to display based on the change.

   Component rendering refers to displaying the correct component based on the selected route. For example, when the route is /add-order, the AddOrderPage component will be rendered in the router-view.

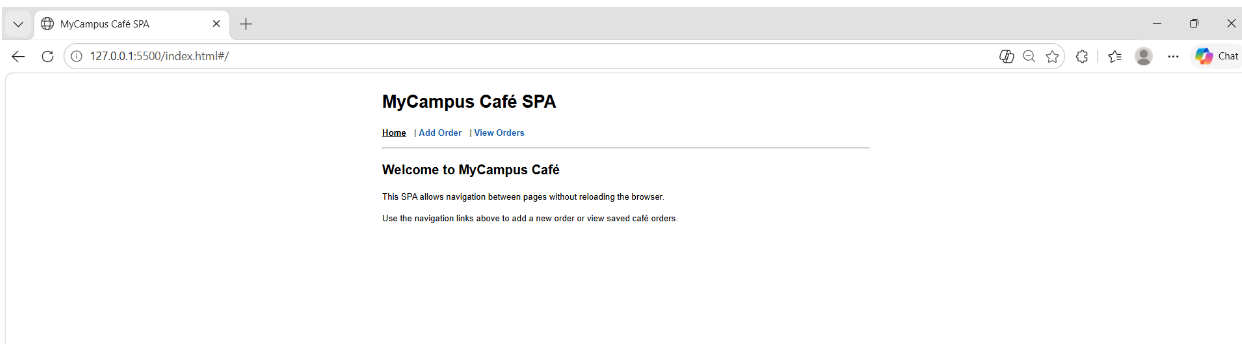


Figure 1.4: Home Page

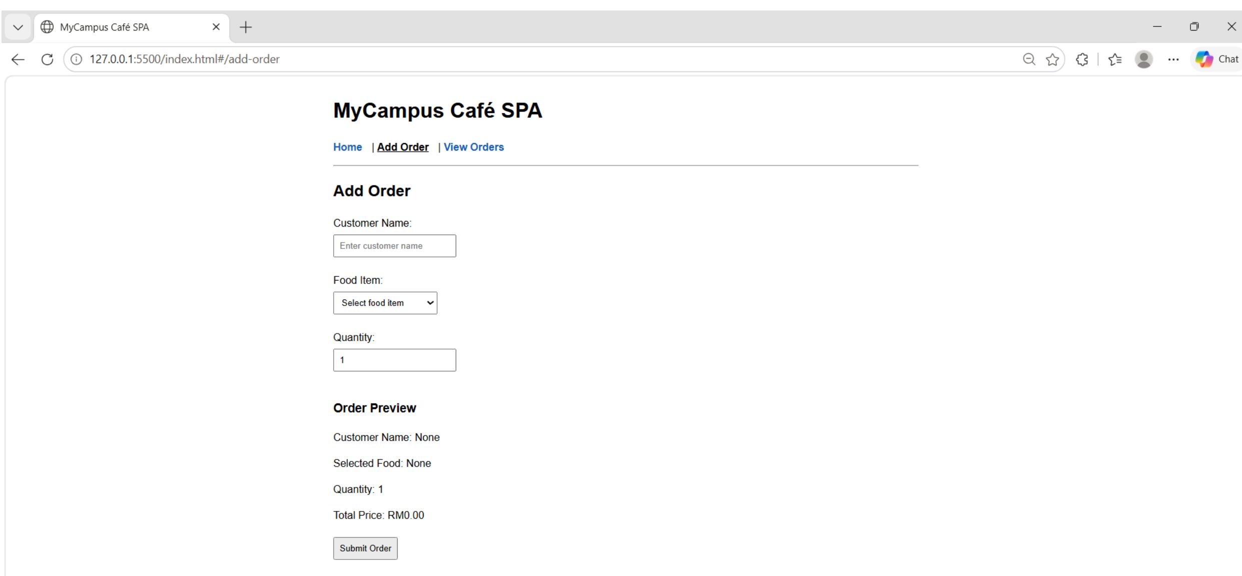


Figure 1.5: Add Order Page

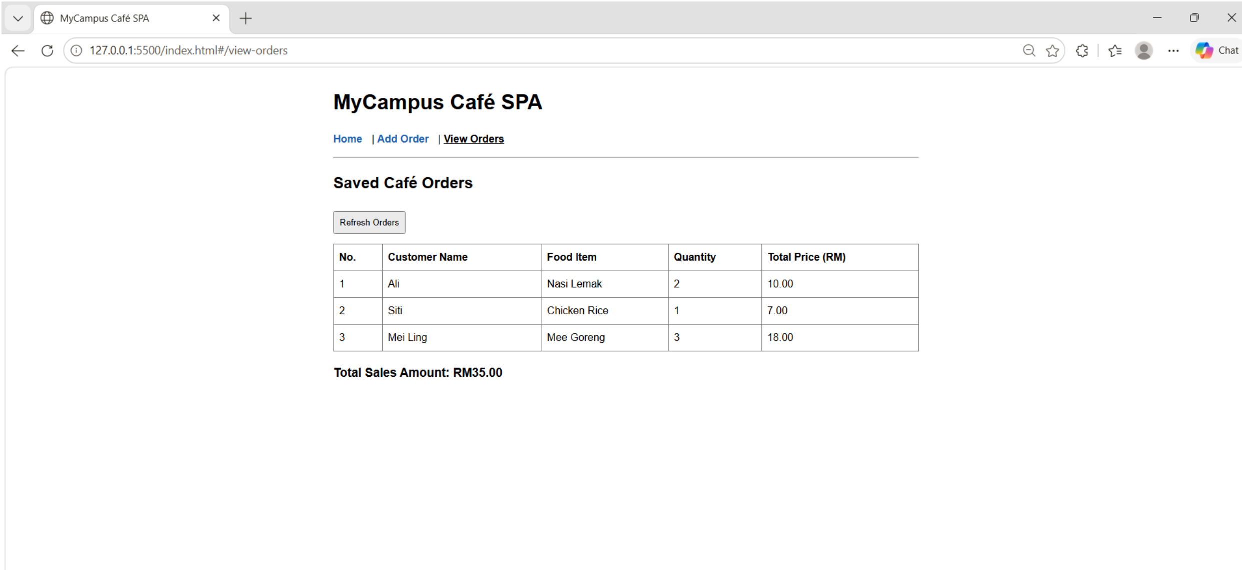


Figure 1.6: View Orders Page